Where does the VAR prior fail?

Block-adaptive Bayesian local projections under sparse dynamic misspecification

MSc thesis project

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The research question

Local projections

Estimate each horizon directly. They tolerate richer dynamics, but impulse responses can be imprecise in modest samples.

VARs

Impose one dynamic system across horizons. They gain precision when that system is adequate, but transmit misspecification everywhere.

A global Bayesian LP compromises by shrinking every coefficient block toward a VAR-implied centre with the same horizon-specific tightness λ_h.

Block-adaptive VAR-centred shrinkage

$$y_{i,t+h} = z_t' \beta_{i,h} + u_{i,t+h},$$

$$\beta_{i,g,h} - \mu_{i,g,h}^{\mathrm{BVAR}} \sim \mathcal{N}\left(0,\ \sigma_{i,h}^2 \lambda_h^2 \tau_{i,g,h}^2 D_{g,h}\right),\quad \tau_{i,g,h} \sim \mathcal{C}^+(0,1).$$

In each response equation and horizon, one block collects all p coefficients attached to the lags of predictor variable g.

$\tau_{i,g,h} < 1$

Tighter than the global prior

$\tau_{i,g,h} = 1$

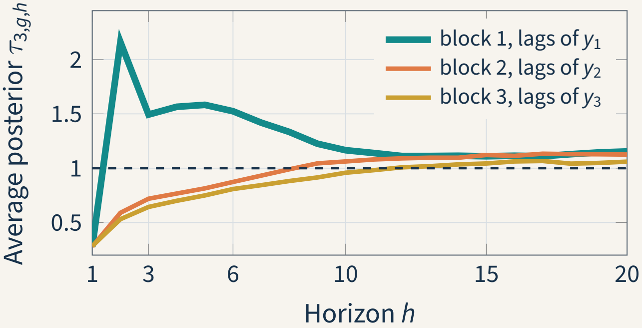
A deliberately diagnostic Monte Carlo design

Regime	True dynamics	What the design tests
Correct	Stable VAR(2)	Whether unnecessary local flexibility costs precision
Sparse target	VAR(3) with only $A_3(3,1) = -0.30$ omitted from the fitted VAR(2)	Whether block 1 can escape in the y_3 equation, where the shock-transmission prior is wrong
Dense fail	VARMA(2,1) with a	Whether the advantage disap

What is implemented now

Method	FMAR BVAR and global BLP baseline, block-adaptive Gibbs sampler, common tightness λ_h, quasi-Bayesian HAC bands.
Nesting	With $\tau = 1$, the adaptive prior recovers the global FMAR posterior-mean map for $h \geq 2$.
Evidence	Committed $R = 500$ results for correct, sparse and dense DGPs in both prototype and FMAR modes.
Verification today	Core Octave tests pass, including prior mapping, analytic true IRFs, sampler checks and FMAR nesting.

The scales locate the failure, but average accuracy does not improve



Sparse DGP, y_3 equation. Monte Carlo average across 500 replications

97.6%

Block 1 ranks largest over $h = 1, \ldots, 6$ in the affected equation.

Correct-DGP reference: 33.0%.

y_3 RMSE	Global	Block
$h = 2$	.1258	**.1041**

What the project can claim today

Established The adaptive estimator is implemented, algebraically nests the FMAR mean for $h \geq 2$, and produces reproducible $R = 500$ simulation results.

Promising The posterior scales identify the deliberately sparse data-prior conflict and correct the targeted response sharply at one early horizon.

Not established The method does not yet lower integrated RMSE, its advantage has not generalized beyond the hand-calibrated sparse design, and it has no empirical validation.

Highest-return next steps

1. **Make the benchmark fair and frozen.** Harmonize $h = 1$ or report the main loss over $h \geq 2$, export stable summary tables, and update the stale project status.
2. **Explain the null result.** Decompose bias and variance by horizon. Vary misspecification size, sample length and fitted lag order to map when local escape can pay.
3. **Pool the scales across horizons.** A smooth prior on $\log \tau_{i,g,h}$ directly targets the current pattern: a strong early signal followed by noisy escape. Add diagnostics for the τ chains.
4. **Use a decision gate.** If pooling does not yield stable risk gains, frame the contribution as a calibrated specification diagnostic.

Restart map

Reorient	Read `README.md` sections 1, 3 and 7–10, then this deck. Treat the README claim that the full FMAR Monte Carlo is pending as stale.
Smoke run	In MATLAB or Octave: `QUICK_DEMO = true; RUN_FMAR_DEMO`.
Test method	Run `cd tests; run_all_tests`. Set `FMAR_PATH` to rerun the optional comparison with the original package.
Inspect evidence	Load `results/mc_fmar_final_{correct,sparse,de` The summary struct `s` contains IRMSE, coverage and